

## SAUMLAKI OLILIT, Indonesia

WMO: 979000

Lat: 7.9825S

Lon: 131.2980E

Elev: 79

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 73.3       | 74.4 | 65.0                          | 92.9 | 77.4 | 66.5 | 97.8 | 78.0 | 22.6                  | 79.5 | 19.7 | 80.3 | 3.6                   | 100  | 0.396 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 11            | 11.7                   | 92.1            | 80.5 | 91.1 | 80.3 | 90.2 | 80.1 | 82.0                | 89.1 | 81.6 | 88.4 | 81.0 | 87.6 | 8.4                  | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.1                            | 156.5 | 86.2 | 79.6 | 154.0 | 85.5 | 79.1 | 151.6 | 84.8 | 46.0          | 89.9 | 45.3 | 88.9 | 44.8 | 88.3 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 19.2          | 16.9 | 15.0 | DB                         | 67.2 | 94.8               | 6.0 | 1.1                                                | 62.9 | 95.6       | 59.4 | 96.2       | 56.0 | 96.8       | 51.6 | 97.6 | (4) |
| (5)               |      |      | WB                         | 64.9 | 85.0               | 3.9 | 1.2                                                | 62.1 | 85.9       | 59.8 | 86.6       | 57.6 | 87.3       | 54.7 | 88.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 82.0   | 83.3  | 83.1  | 82.9  | 82.7  | 82.2  | 80.4  | 79.7  | 79.6  | 80.6  | 82.4  | 84.0  | 83.7 |
|                                                                                  | DBStd       | 2.35   | 2.17  | 2.06  | 2.07  | 1.98  | 1.81  | 1.77  | 1.50  | 1.34  | 1.55  | 1.64  | 1.54  | 2.09 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 11690  | 1031  | 926   | 1021  | 982   | 997   | 911   | 920   | 917   | 917   | 1005  | 1019  | 1044 |
|                                                                                  | CDD65       | 6215   | 566   | 506   | 556   | 532   | 532   | 461   | 455   | 452   | 467   | 540   | 569   | 579  |
|                                                                                  | CDH74       | 65577  | 6310  | 5613  | 6112  | 5963  | 5742  | 4317  | 3852  | 3753  | 4435  | 5978  | 6858  | 6644 |
|                                                                                  | CDH80       | 19952  | 2171  | 1908  | 2065  | 2004  | 1626  | 736   | 504   | 567   | 1050  | 2006  | 2822  | 2493 |
| Wind                                                                             | WSAvg       | 6.9    | 6.5   | 6.7   | 5.1   | 5.3   | 7.9   | 9.1   | 9.6   | 9.1   | 7.6   | 6.6   | 4.7   | 4.8  |
| Precipitation                                                                    | PrecAvg     | 44.5   | 5.5   | 5.8   | 6.3   | 5.8   | 6.9   | 4.0   | 0.9   | 0.5   | 0.1   | 1.0   | 1.7   | 6.1  |
|                                                                                  | PrecMax     | 118.3  | 16.3  | 17.6  | 17.5  | 24.4  | 25.7  | 21.8  | 3.7   | 3.6   | 1.3   | 13.7  | 7.8   | 18.5 |
|                                                                                  | PrecMin     | 2.5    | 0.0   | 0.1   | 0.0   | 0.3   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 30.1   | 5.8   | 5.6   | 5.9   | 6.8   | 7.4   | 5.5   | 1.1   | 0.8   | 0.2   | 2.7   | 2.3   | 5.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 92.5  | 92.3  | 92.1  | 90.9  | 89.1  | 87.6  | 85.5  | 85.7  | 88.0  | 90.6  | 93.1  | 93.6 |
|                                                                                  |             | MCWB   | 81.2  | 81.3  | 81.4  | 80.5  | 80.2  | 78.5  | 76.0  | 76.4  | 78.1  | 78.7  | 80.2  | 80.5 |
|                                                                                  | 2%          | DB     | 90.9  | 90.7  | 90.7  | 89.6  | 87.9  | 85.9  | 84.5  | 84.7  | 86.7  | 89.0  | 91.6  | 92.1 |
|                                                                                  |             | MCWB   | 80.8  | 81.0  | 80.9  | 80.0  | 79.2  | 77.6  | 76.1  | 75.9  | 77.4  | 78.4  | 79.7  | 80.3 |
|                                                                                  | 5%          | DB     | 89.7  | 89.4  | 89.4  | 88.6  | 87.0  | 84.8  | 83.6  | 83.8  | 85.8  | 88.1  | 90.5  | 90.7 |
|                                                                                  |             | MCWB   | 80.6  | 80.6  | 80.5  | 79.7  | 78.7  | 77.1  | 75.8  | 75.4  | 76.9  | 78.2  | 79.3  | 80.1 |
|                                                                                  | 10%         | DB     | 88.3  | 88.1  | 88.2  | 87.5  | 86.0  | 83.7  | 82.8  | 83.0  | 84.8  | 87.2  | 89.4  | 89.1 |
|                                                                                  |             | MCWB   | 80.3  | 80.2  | 80.2  | 79.4  | 78.3  | 76.7  | 75.5  | 75.0  | 76.3  | 77.8  | 78.9  | 80.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 82.5  | 82.7  | 82.7  | 82.0  | 81.7  | 79.8  | 78.9  | 78.3  | 79.8  | 80.3  | 81.9  | 82.8 |
|                                                                                  |             | MCDB   | 89.7  | 89.4  | 89.8  | 88.5  | 87.4  | 85.0  | 82.9  | 83.3  | 86.1  | 87.3  | 90.4  | 90.5 |
|                                                                                  | 2%          | WB     | 81.6  | 81.7  | 81.7  | 81.2  | 80.6  | 78.9  | 78.0  | 77.4  | 78.8  | 79.6  | 80.8  | 81.6 |
|                                                                                  |             | MCDB   | 88.5  | 88.6  | 88.5  | 87.5  | 86.4  | 83.8  | 82.2  | 82.5  | 84.6  | 86.6  | 89.0  | 89.0 |
|                                                                                  | 5%          | WB     | 81.0  | 81.0  | 81.0  | 80.6  | 79.9  | 78.2  | 77.3  | 76.6  | 77.9  | 79.0  | 80.1  | 80.8 |
|                                                                                  |             | MCDB   | 87.4  | 87.5  | 87.3  | 86.8  | 85.2  | 82.8  | 81.5  | 81.8  | 83.8  | 86.0  | 88.0  | 87.8 |
|                                                                                  | 10%         | WB     | 80.5  | 80.5  | 80.5  | 79.9  | 79.2  | 77.6  | 76.6  | 76.0  | 77.2  | 78.5  | 79.6  | 80.2 |
|                                                                                  |             | MCDB   | 86.6  | 86.6  | 86.5  | 85.6  | 84.1  | 82.0  | 81.0  | 81.2  | 83.1  | 85.3  | 87.2  | 86.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.6   | 9.6   | 9.8   | 9.0   | 7.4   | 6.4   | 6.6   | 7.4   | 8.9   | 10.1  | 11.7  | 10.8 |
|                                                                                  |             | MCDBR  | 11.5  | 11.5  | 11.6  | 10.7  | 8.6   | 7.8   | 7.4   | 8.0   | 9.7   | 10.8  | 13.0  | 13.2 |
|                                                                                  |             | MCWBR  | 4.2   | 4.3   | 4.4   | 4.5   | 4.0   | 3.6   | 3.7   | 3.8   | 3.7   | 3.8   | 4.3   | 4.3  |
|                                                                                  | 5% WB       | MCDBR  | 10.5  | 10.4  | 10.8  | 9.6   | 8.1   | 6.8   | 6.5   | 7.4   | 8.7   | 10.0  | 12.1  | 11.6 |
|                                                                                  |             | MCWBR  | 4.4   | 4.5   | 4.6   | 4.4   | 4.1   | 3.6   | 3.6   | 4.0   | 3.8   | 3.9   | 4.4   | 4.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.424  | 0.420 | 0.405 | 0.406 | 0.430 | 0.432 | 0.418 | 0.413 | 0.421 | 0.442 | 0.431 | 0.419 |      |
|                                                                                  | taud        | 2.481  | 2.484 | 2.532 | 2.532 | 2.452 | 2.419 | 2.462 | 2.465 | 2.419 | 2.350 | 2.412 | 2.477 |      |
|                                                                                  | Ebn at Noon | 291    | 291   | 291   | 282   | 266   | 260   | 267   | 276   | 282   | 281   | 287   | 292   |      |
|                                                                                  | Edh at Noon | 37     | 37    | 35    | 33    | 34    | 35    | 34    | 35    | 38    | 42    | 39    | 37    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1614   | 1701  | 1682  | 1651  | 1511  | 1413  | 1537  | 1840  | 2074  | 2147  | 2088  | 1737  |      |
|                                                                                  | RadStd      | 194    | 183   | 146   | 170   | 164   | 140   | 94    | 92    | 96    | 106   | 118   | 194   |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page